

03 · Price elasticity by region — random slopes & endogeneity

July 12, 2026

The business decision. We set prices region by region and want each region’s **price elasticity of demand** — the standard economics measure of how sensitive demand is to price. Concretely, elasticity is the **% change in quantity sold for a 1% change in price**; it’s almost always negative (raise price -> sell less). An elasticity of -1.5 means “cut price 1%, sell 1.5% more.” We need it per region because a tourist market and a price-sensitive suburb should not be priced the same.

0.0.1 Two traps this notebook is built around

1. **Small regions are noisy.** A region with 12 weeks of data will hand you a wild elasticity estimate that’s mostly sampling noise. Acting on it is overfitting. The fix is **partial pooling** (below).
2. **Price is endogenous.** “Endogenous” means the treatment (price) is *correlated with the error term* — here because managers historically *cut prices when they expected soft demand*. So low prices coincide with low demand for reasons that have nothing to do with the price cut, which drags the naive elasticity estimate *toward zero* (makes demand look less price-sensitive than it is). No amount of clever averaging fixes this — it needs an **instrument** (notebook 11), which we preview at the end.

0.0.2 The tools, in plain terms

- **Log-log demand.** We model $\log(\text{quantity}) = \alpha + \beta \log(\text{price}) + \dots$. The magic of taking logs on both sides is that the slope β **is** the elasticity directly (a constant %-for-% relationship), so there’s one interpretable number per region. One line of calculus shows why: elasticity is defined as $\epsilon(P) \equiv \frac{dQ}{dP} \cdot \frac{P}{Q}$, and for the constant-elasticity curve $Q = AP^\beta$ we get $\epsilon(P) = \beta AP^{\beta-1} \cdot \frac{P}{AP^\beta} = \beta$ — the *same* β at every price point. For **any other** demand shape the elasticity moves with the price level, so β is best read as a *local approximation around the prices we actually observed* — a point that returns with teeth when we recommend prices outside that range.
- **Random slopes / partial pooling.** Instead of one elasticity for all regions (ignores real differences) or a separate free estimate per region (noisy), we assume each region’s elasticity β_r is drawn from a shared distribution $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta)$. This **hierarchical** structure makes each region’s estimate a compromise between its own noisy data and the fleet average — data-poor regions get pulled (“shrunk”) harder toward the fleet, which is exactly the protection we want. μ_β is the typical elasticity; τ_β measures how much regions genuinely differ.

Cookbook maps this to **pathmc** random slopes; here we fit the equivalent hierarchical model explicitly in PyMC so the shrinkage is fully visible. The identification logic is unchanged.

On real data. Swap in your **own transaction panel** — one row per region per week with price, units sold, and controls (competitor price, promotions, seasonality). Public analogues are retail *scanner* datasets (e.g. the Dominick’s Finer Foods store data). The endogeneity caveat is not academic: it’s the single biggest reason naive “price tests” mislead in practice.

0.1 2 · Simulate a ground truth

A region x week panel with a **constant-elasticity (log-log) demand curve** $\log Q_{rt} = \alpha_r + \beta_r \log P_{rt} + \gamma^\top Z_{rt} + \varepsilon_{rt}$, where β_r is region r ’s true price elasticity (planted, varying around a fleet mean ~ -1.4). Of the observed covariates Z , only **competitor price** is a genuine confounder — it moves both our price and demand; **season** and **trend** shift demand only, so they enter as *precision covariates* (they sharpen estimates), not confounders. We start with **no** endogeneity so we can show recovery, then switch it on in step 8.

The data-generating model — exactly what `dgp.price_panel` implements (defaults & seed in `src/cmp/dgp.py`). For region r , week t : season $s_t = 0.3 \sin(2\pi t/26)$, trend $g_t = 0.01 t$, competitor price $C_{rt} \sim \mathcal{N}(20, 2^2)$, and a demand shock $u_{rt} \sim \mathcal{N}(0, 1)$:

$$\begin{aligned} \beta_r &\sim \mathcal{N}(-1.4, 0.35^2) \\ \log P_{rt} &= \log 15 - 0.05 C_{rt}/20 + \kappa u_{rt} + \nu_{rt}, \quad \nu_{rt} \sim \mathcal{N}(0, 0.05^2) \\ \log Q_{rt} &= 2.5 + \beta_r \log P_{rt} + 0.4 \frac{C_{rt}-20}{20} + s_t + g_t + u_{rt} + \varepsilon_{rt}, \quad \varepsilon_{rt} \sim \mathcal{N}(0, 0.08^2). \end{aligned}$$

Line 1 plants the true regional elasticities; line 2 is the price-setting rule; line 3 is demand. C_{rt} appears in **both** equations — that is what makes competitor price the genuine confounder (it moves our price *and* demand), while s_t and g_t appear only in the demand equation (precision covariates). κ is the **endogeneity switch**: through step 6 the generator runs with $\kappa = 0$ and zeroes the shock entirely ($u_{rt} \equiv 0$), so recovery is clean; step 8 sets $\kappa > 0$, and the *same* shock now enters both equations — managers’ prices respond to the very shock that moves demand ($\text{Cov}(\log P, u) > 0$), the simultaneity that biases the naive elasticity toward zero.

12 regions, fleet-mean elasticity -1.42; obs per region range 16-80 (regions 0-2 are thin, to exercise shrinkage)

	region	week	price	demand	competitor_price	season \
0	region_00	2	13.003239	0.159630	20.405765	1.394170e-01
1	region_00	3	14.245949	0.157405	16.535730	1.989368e-01
2	region_00	12	14.310937	0.159097	20.819276	7.179470e-02
3	region_00	13	15.626376	0.175255	21.659711	-9.648736e-17
4	region_00	15	13.204126	0.152195	19.486540	-1.394170e-01
	trend	log_price	log_demand			
0	0.02	2.565199	-1.834900			
1	0.03	2.656473	-1.848934			
2	0.12	2.661024	-1.838239			
3	0.13	2.748960	-1.741513			
4	0.15	2.580529	-1.882592			

0.2 3 · Identify — what must be true for β_r to be causal?

Identification vs estimation. *Identification* asks whether the causal quantity could be recovered from this data even with infinite samples — a question about **assumptions**, settled before any fitting; *estimation* is the statistics we do once identification is granted. Partial pooling (below) is an **estimation** device — it buys variance reduction, never identification.

Estimand. Region r 's elasticity is a *causal* slope — what happens to demand when **we** move the price, not when the market moves it for us:

$$\beta_r = \frac{\partial \mathbb{E}[\log Q \mid \text{do}(\log P = p), r]}{\partial p},$$

where $\text{do}(\cdot)$ marks an *intervention* on price (Pearl's do-operator), as opposed to passively observing whatever prices managers happened to choose. The regression slope equals this causal slope only under assumptions — four of them, each named, formalized, and honestly labelled as checkable or not:

A1 · Conditional price exogeneity (the continuous-treatment version of unconfoundedness):

$$\varepsilon_{rt} \perp \log P_{rt} \mid Z_{rt}, r.$$

After conditioning on region and the observed controls Z , whatever noise remains in demand must be unrelated to how the price was set. The one true observable confounder in our DGP is **competitor price** — it moves both P and Q , opening the backdoor path in the DAG below — and conditioning on it closes that path. **Season** and **trend** move demand only, so they are *precision covariates* (they sharpen estimates), not confounders. A1 holds by construction while $\kappa = 0$; it is *exactly* the assumption we break in step 8, and it is **untestable from the data alone** — no diagnostic in this notebook can certify it.

A2 · Price variation (positivity, continuous-treatment version):

$$\text{Var}(\log P_{rt} \mid Z_{rt}, r) > 0.$$

Each region must have genuine price movement left over after the controls; no variation, no slope. In the DGP this is the quasi-experimental wiggle $\nu_{rt} \sim \mathcal{N}(0, 0.05^2)$ — checked numerically in the next cell. In real data this is the assumption most often *silently* violated: uniform corporate pricing or rigid cost-plus rules leave $\text{Var}(\log P \mid Z) \approx 0$, and the “elasticity” you then estimate is noise dressed up as a number.

A3 · No dynamic effects / SUTVA. This week's price affects this week's demand only: no stockpiling (“pantry-loading” ahead of a price rise), no delayed reference-price effects, and no cross-region spillovers (region A's price doesn't move region B's demand — no arbitrage). Our DGP satisfies this by construction; on real scanner data a **lag(price)** term is the standard first probe (step 8's caveats).

A4 · Functional form. In a *parametric* model the constant-elasticity (log-log) shape is part of the identification bundle: if the true curve bends, the “constant” β_r is a price-range-specific average. We stress-test this in-sample in step 5 with a curvature term — but no in-sample check reaches prices we never charged.

Partial pooling, and what it buys. The estimand is per-region, so we treat elasticity as a **random slope**: $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta)$ — noisy regions shrink toward the fleet mean μ_β , and τ_β measures how much elasticity genuinely varies. The **shrinkage factor** for region r is approximately $1 - \frac{\tau_\beta^2}{\tau_\beta^2 + \sigma^2/n_r}$ — thin regions (small n_r) get pulled harder toward the fleet. Shrinkage is a feature: it stops us over-reacting to a small region’s noisy slope. But pooling handles *noise*, not *bias*: if A1 fails, all three pooling regimes in step 4 are biased by the same amount.

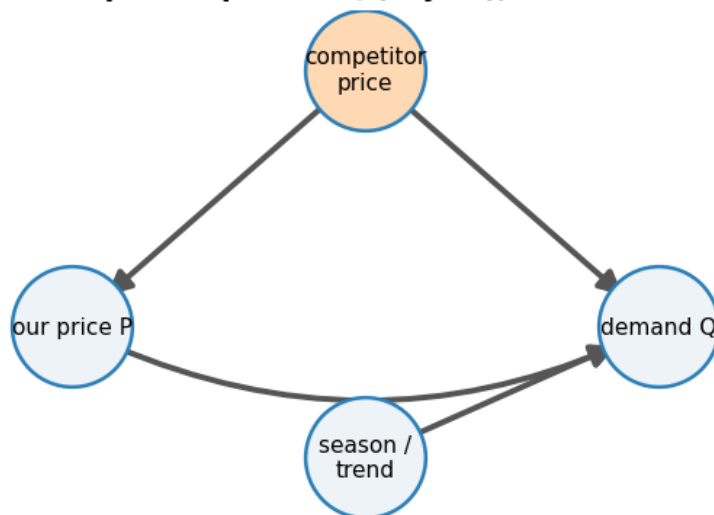
How big is the bias if A1 fails? Two lines of algebra, not hand-waving. Suppose an unobserved shock u_{rt} (variance σ_u^2) enters demand with coefficient 1 *and* leaks into price with coefficient κ (managers cut price when they expect soft demand). For the regression of $\log Q$ on $\log P$ after partialling out Z and region effects, the classic **omitted-variable-bias** formula gives

$$\hat{\beta} \xrightarrow{p} \beta + \frac{\text{Cov}(\log P, u)}{\text{Var}(\log P)} = \beta + \frac{\kappa \sigma_u^2}{\kappa^2 \sigma_u^2 + \sigma_\nu^2},$$

using the DGP’s price equation ($\text{Cov}(\log P, u) = \kappa \sigma_u^2$ and $\text{Var}(\log P \mid Z) = \kappa^2 \sigma_u^2 + \sigma_\nu^2$). The bias term is **positive**, and since $\beta < 0$ it drags $\hat{\beta}$ **toward zero** — demand looks less price-sensitive than it is. Worse, the *sign of the whole slope can flip*: when the exogenous wiggle σ_ν is small, even a faint κ makes the bias enormous (the denominator is tiny) and the fitted “demand curve” slopes *upward* — high prices coinciding with high demand because both chase the same shock. Step 8 verifies this arithmetic numerically, then sweeps κ to map the whole curve. What pooling does **not** fix is exactly this: an *unobserved* common shock needs an instrument — previewed in step 8, full treatment in notebook 11. We state this out loud rather than hide it.

A2 check - within-region SD of log-price: min 0.046, median 0.052, max 0.058
(all comfortably > 0 : positivity holds, by design)

Backdoor P ← competitor price → Q (adjust); season/trend = precision



0.3 4 · Estimate — three pooling regimes, side by side

To make the bias–variance trade-off concrete we estimate each region’s elasticity **three ways**:

- **No pooling** — a separate regression per region. Unbiased, but a data-poor region’s estimate is pure noise. This is the “trust each region’s own numbers” extreme.
- **Complete pooling** — one elasticity for the whole fleet. Rock-steady, but pretends every region is identical, so it’s biased for any region that genuinely differs. The “everyone’s the same” extreme.
- **Partial pooling** — the hierarchical Bayesian middle ground: each β_r is nudged toward the fleet mean by an amount that depends on how much data the region has. Data-rich regions barely move; data-poor regions borrow heavily. This is the one that gets *both* the overall level and the spread right.

The model below reads directly as the maths from Step 3: `mu_beta` is the fleet-mean elasticity μ_β , `tau_beta` is how much regions differ τ_β , and `beta_r` are the per-region elasticities drawn from $\mathcal{N}(\mu_\beta, \tau_\beta)$. `alpha_r` are region baselines and `gamma` the controls (competitor price, seasonality, trend). The printout compares the recovered fleet mean to the planted truth.

The fitted model, in symbols (observation i in region $r[i]$; z_i the standardized controls — competitor price, season, trend):

$$\begin{aligned} \log Q_i &\sim \mathcal{N}(\alpha_{r[i]} + \beta_{r[i]} \log P_i + \gamma^\top z_i, \sigma^2) \\ \beta_r &\sim \mathcal{N}(\mu_\beta, \tau_\beta^2), \quad \alpha_r \sim \mathcal{N}(0, 5^2), \quad \gamma \sim \mathcal{N}(0, 2^2) \\ \mu_\beta &\sim \mathcal{N}(-1, 1^2), \quad \tau_\beta \sim \text{HalfNormal}(1), \quad \sigma \sim \text{HalfNormal}(1). \end{aligned}$$

The middle line’s $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$ is the partial pooling: τ_β (how much regions truly differ) is *learned from the data*, and each region’s posterior elasticity is a precision-weighted compromise between its own weeks and the fleet mean — thin regions lean on the fleet, data-rich regions stand on their own.

Why these priors. $\mu_\beta \sim \mathcal{N}(-1, 1)$ puts ~95% of its mass on $(-3, +1)$: it spans everything from mildly inelastic demand to $\beta \approx -3$ (more price-sensitive than almost any retail elasticity ever measured) while gently centering on the economically plausible region — and it is weak enough that the panel dominates it easily. $\tau_\beta \sim \text{HalfNormal}(1)$ lets the data choose the **pooling regime itself**: $\tau_\beta \rightarrow 0$ collapses to complete pooling (all regions identical), large τ_β approaches no pooling (every region on its own) — the two baselines above are the *limiting cases* of the model we actually fit, and where we land between them is *learned*, not assumed. $\alpha_r \sim \mathcal{N}(0, 5)$ is diffuse for log-scale intercepts, $\gamma \sim \mathcal{N}(0, 2)$ is weakly informative for standardized controls, and $\sigma \sim \text{HalfNormal}(1)$ is generous for log-demand residual noise (the DGP’s is 0.08).

A practitioner’s note on parametrization. We fit the *centered* form $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$, and here it samples cleanly (check the divergence count in the convergence report below). But centered hierarchies are *the* canonical NUTS pitfall: with few groups or weak data the joint posterior of (τ_β, β_r) develops a **funnel** — as $\tau_\beta \rightarrow 0$ the β_r are squeezed into an ever-narrower band the sampler cannot enter. The standard fix is the **non-centered** form $\beta_r = \mu_\beta + \tau_\beta \tilde{\beta}_r$ with $\tilde{\beta}_r \sim \mathcal{N}(0, 1)$, which decorrelates the geometry. If you port this notebook to a panel with 4 regions and 12 weeks, switch to non-centered first.

Like-for-like baselines. The no-pool and complete-pool baselines are fit with the **same standardized controls** Z (competitor price, season, trend) as the hierarchical model — per-region OLS and pooled OLS with region intercepts, respectively — so the MAE comparison in step 6 isolates the **pooling choice** and nothing else. (An earlier draft used bivariate log Q -on-log P fits for the baselines, which conflated the pooling choice with the control choice; clean comparisons are four lines of code, so we make them.)

```
hierarchical model convergence: max r-hat 1.010 - min ESS 251 - divergences 0
fleet-mean elasticity mu_beta -1.36 (true -1.42) · between-region spread
tau_beta 0.31 (true 0.31)
```

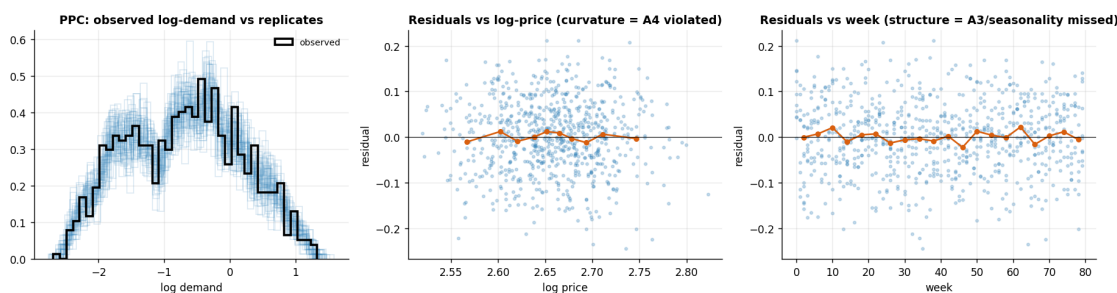
0.4 5 · Check the model *before* grading it — posterior predictive & functional form

Discipline first: before we congratulate the model on recovering the planted truth (a luxury a real analyst never has), we run the checks a real analyst *can* run.

- **Posterior predictive check (PPC).** Simulate replicated datasets $\log Q^{\text{rep}}$ from the fitted model and overlay their distribution on the observed $\log Q$. If the cloud of replicates fails to cover the data — wrong location, spread, or shape — the model is structurally wrong somewhere.
- **Residuals against log-price and against week.** Curvature in residuals-vs-log P is exactly what a violated constant-elasticity assumption (A4) looks like; leftover structure in residuals-vs-week means missed seasonality or dynamics (A3).
- **A curvature term as a formal probe of A4.** We add $(\log P - \overline{\log P})^2$ to a pooled OLS with the same controls and check whether its coefficient is distinguishable from zero.

And the sentence that matters most: a clean PPC certifies **fit, not causality**. The endogenous model of step 8 — wrong by nearly a full elasticity point — would sail through this same PPC, because confounding is invisible in the distribution of the outcome alone. PPC failures tell you the model is wrong; PPC successes do **not** tell you it is right about the causal question. That was assumption A1, and A1 is untestable. Keep the two firmly apart.

A4 curvature probe - coefficient on $(\log P - \text{mean})^2$: -0.9 (SE 0.8, $t = -1.0$) -> consistent with zero -> constant elasticity survives in-sample
note the wide SE: with only +/-5% price wiggle this test has little power far from current prices -
another reason to treat any P^* outside the observed range as extrapolation, not measurement.
PPC replicate mean -0.70 vs observed -0.70; replicate SD 0.88 vs observed 0.88



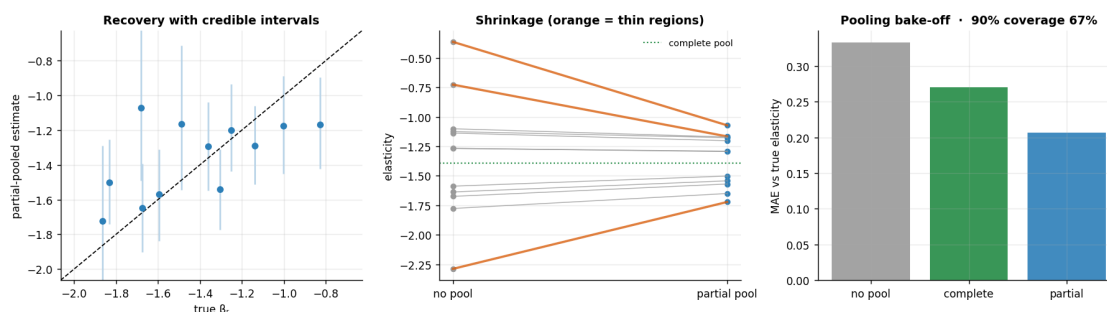
How to read the three panels. *Left* — the black observed log-demand density sits inside the blue cloud of posterior replicates: the model reproduces the location, spread and shape of the outcome. *Middle* — residuals vs log-price show no systematic bend: the binned means (orange) hug the zero line across the observed price range, so constant elasticity is consistent with the data *where we have data*. *Right* — residuals vs week show no leftover seasonal wave or drift; the sine/trend controls are doing their job. The curvature probe printed under the figure makes the middle panel formal — and its wide standard error is itself informative: with only $\pm 5\%$ of price variation, in-sample data can never certify the demand shape far from current prices. **None of this validates A1 (exogeneity):** the step-8 model passes every one of these checks while being badly wrong about the causal slope.

0.5 6 · Validate — recovery, shrinkage, and calibration

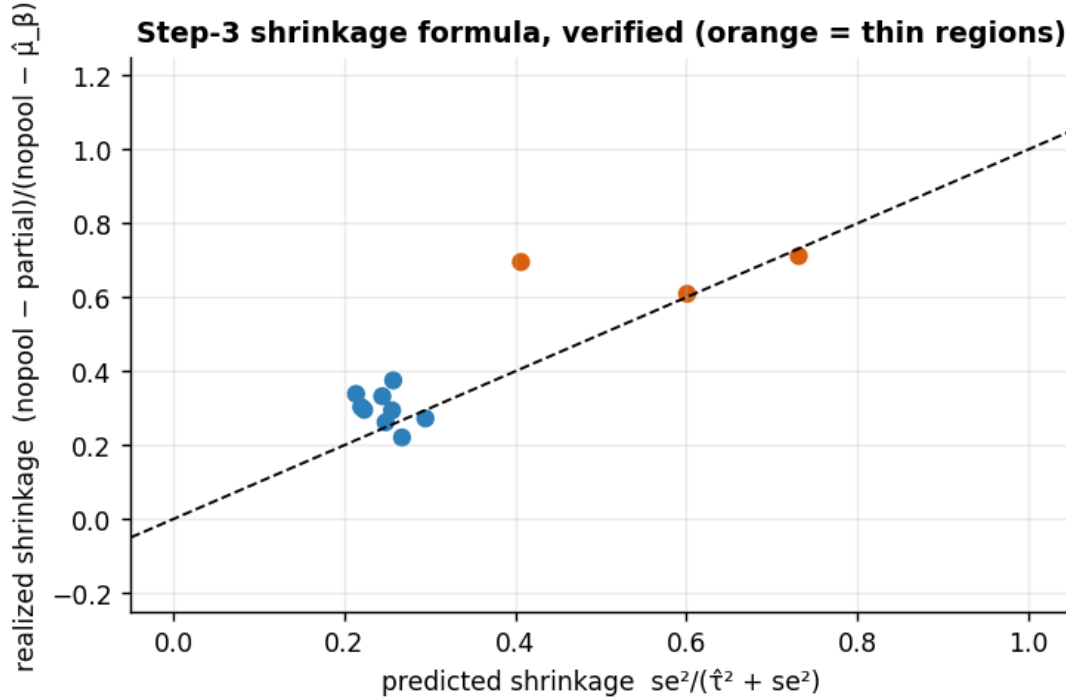
Estimated vs true elasticity should track the 45° line; the thin regions' point estimates get pulled toward the fleet mean (that's pooling protecting us), and the credible intervals are wider where data is thin. We also check that the 90% intervals actually **cover** the true elasticities (calibration).

MAE - no pool 0.33, complete 0.27, partial 0.21 · coverage 67%

coverage split - thin regions (0-2): 2/3 covered · rich regions: 6/9 covered
with only 12 regions, a perfectly calibrated 90% interval covers 8 or fewer with probability 0.03 -> a real shortfall, not just noise



thin regions - predicted shrinkage [0.73 0.6 0.41], realized [0.71 0.61 0.7]
rich regions - predicted 0.25 on average, realized 0.30 (0 region(s) hidden: no-pool slope already ~ fleet mean, so the ratio is ill-defined)



How to read those panels. *Left* — partial-pooled estimates track the true elasticities along the 45° line, and the error bars are visibly wider for the thin regions (less data -> more honest uncertainty). *Middle* — the shrinkage in action: lines connect each region’s noisy no-pool estimate (left) to its partial-pooled estimate (right); the **thin regions (orange) get yanked hardest** toward the fleet, while data-rich regions barely budge. That selective pull is exactly the protection partial pooling buys. *Right* — the bottom line: mean absolute error against the truth is **lowest for partial pooling**, beating both extremes *even now that the baselines get the same controls* — no-pooling is noisy, complete-pooling is biased, and the remaining gap is the pooling choice alone.

Is the below-nominal coverage a bug? Don’t wave it away — check it, twice; both checks are printed under the figure. The **split** locates any misses: shrinkage intervals are calibrated for the *ensemble*, not for each pre-shrinkage slope, so intervals run too narrow precisely for regions whose true elasticity sits far from the fleet mean and for the hard-shrunk thin ones. The **binomial line** asks whether the shortfall is even signal: with 12 regions, a perfectly calibrated 90% procedure still misses a couple regularly, and the printed probability says on which side of that line this run falls. When it flags a *real* shortfall, that is the known price of shrinkage — slightly overconfident **per-region** intervals bought in exchange for the best **point** estimates (lowest MAE, right panel) — worth knowing before quoting an interval to a region manager, and one more reason the decision rules in step 7 lean on $P(\beta_r < -1)$ across the whole posterior rather than on interval endpoints. The follow-up scatter then closes the loop on step 3’s shrinkage formula: the predicted pull $se_r^2/(\hat{\tau}_\beta^2 + se_r^2)$ tracks the pull the posterior actually applied (45° line), thin regions (orange) toward the heavy-shrinkage corner — the formula we asked you to trust in step 3 is the formula the sampler obeyed.

0.6 7 · Decide, in euros — profit-maximising price per region

The decision-maker’s question is not “what is β ?” but “**what price do I set?**” For constant-elasticity demand $Q(P) = AP^\beta$ and constant marginal cost c , weekly profit is

$$\pi(P) = (P - c)Q(P) = (P - c)AP^\beta.$$

Differentiate and set to zero:

$$\frac{d\pi}{dP} = AP^\beta + (P - c)\beta AP^{\beta-1} = AP^{\beta-1}[(\beta + 1)P - \beta c] = 0 \implies P^* = \frac{\beta}{\beta + 1}c,$$

valid **only when** $\beta < -1$ (demand elastic). The bracket $(\beta + 1)P - \beta c$ then starts positive at small P and decreases through zero — profit rises, peaks at P^* , falls: a genuine maximum (that sign change *is* the second-order condition). For $-1 \leq \beta < 0$ the bracket stays positive for every $P > 0$, so the model says “raise price without limit” — which really means *the constant-elasticity model has left its domain of validity*, not that infinite prices are optimal.

Rearranged, P^* is the classic **Lerner index** (markup) rule:

$$\frac{P^* - c}{P^*} = -\frac{1}{\beta}$$

— *optimal markup over cost equals the inverse elasticity*, the oldest one-liner in pricing: $\beta = -2$ prices at twice cost, $\beta = -1.5$ at three times cost. And note the **blow-up as** $\beta \rightarrow -1^-$: the denominator $\beta + 1 \rightarrow 0^-$ and $P^* \rightarrow +\infty$. Posterior draws near -1 therefore produce wild optimal prices — the statistical reason (exploited below) why regions whose elasticity posterior straddles -1 get routed to a **controlled test**, not a price move.

We propagate the **elasticity posterior** draw-by-draw into a **price posterior** and a **profit-uplift posterior** per region, and — the honest part — flag regions where we’re not confident demand is even elastic ($\beta < -1$) for that test rather than acting on a shaky slope.

region	elasticity	P(elastic)	P* €	P* 90% CI	uplift %	action
region_00	-1.07	0.63	-	-	35	controlled test
region_01	-1.72	1.00	19.6	[14.3, 35.6]	12	set price
region_02	-1.17	0.75	-	-	35	controlled test
region_03	-1.20	0.88	-	-	44	controlled test
region_04	-1.18	0.85	-	-	43	controlled test
region_05	-1.29	0.97	34.4	[22.6, 109.8]	36	set price
region_06	-1.57	1.00	22.2	[17.6, 33.7]	16	set price
region_07	-1.65	1.00	20.3	[16.9, 28.3]	11	set price
region_08	-1.29	0.98	35.1	[23.6, 100.9]	37	set price
region_09	-1.17	0.85	-	-	45	controlled test
region_10	-1.54	1.00	22.7	[18.3, 33.8]	17	set price
region_11	-1.50	1.00	24.0	[18.8, 39.4]	19	set price

Across the 7 confidently-elastic regions, moving to optimal pricing lifts profit

by +21%

[90% +13%, +30%] on average - the pricing change's value, scale-free and with uncertainty carried through (assumes constant elasticity out to P^* ; see caveats).

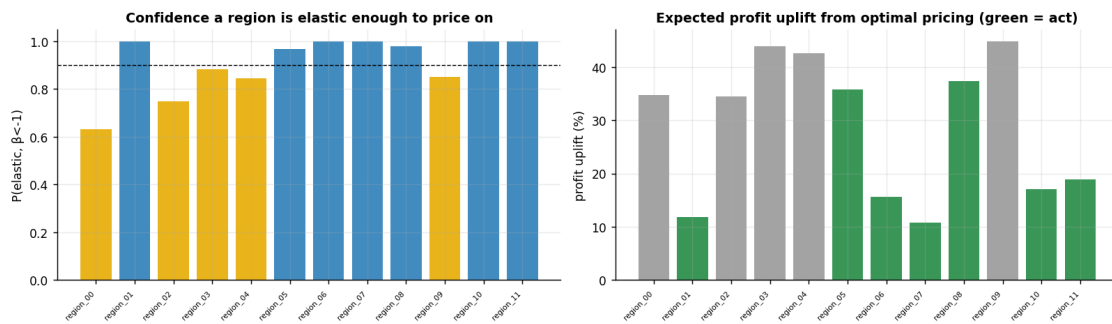
point-plug illustration - weakest region region_00: $P(\text{elastic}) = 0.63$, mean beta = -1.07;

plugging that mean into $P^* = c\beta/(\beta+1)$ gives €121, vs €20 for the most confident region (region_01).

its expected uplift +35% is diluted by the 37% of draws pinned at 0%;

conditional on truly

being elastic it is +55% - a big, **unreliable** number, which is exactly why it goes to a test.

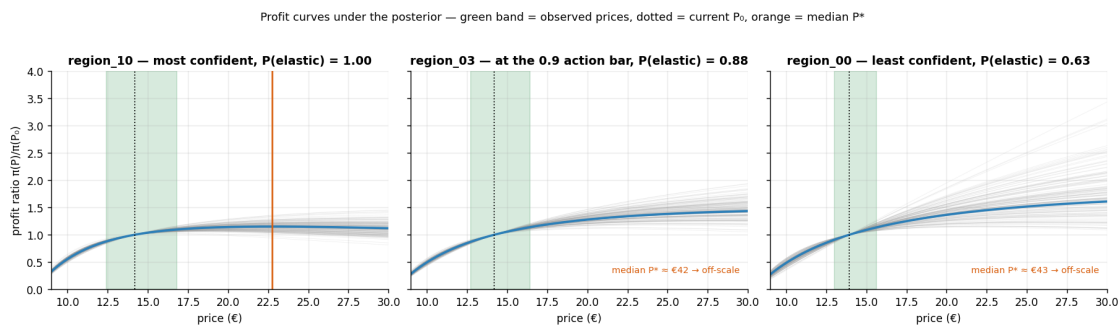


Green band = prices ever charged; the model was only ever tested there (step 5).

Both P_0 (dotted) and the

median P^* (orange) matter less than the SHAPE: a flat top means a mis-estimated beta costs little profit;

curves that fan out and keep rising off the right edge mean the 'optimum' is an extrapolated guess.



How to read the decision table and the two panels. *Left* — each bar is $P(\beta < -1)$, our posterior confidence that a region's demand is **elastic** enough to price on: **blue** clears the 0.9 line (*set price*), **gold** sits in 0.5–0.9 (*controlled test*), **orange** is below 0.5 (*hold/test*). *Right* — the

expected **profit uplift** from moving to the optimal price, **green** only where we are confident enough to act. The fleet number underneath averages the uplift across the confidently-elastic regions and carries the posterior through to a 90% interval. The **P*** columns are the number the CMO actually signs: the posterior-median optimal price with its 90% credible interval — printed *only* for regions confident enough to act on (“—” elsewhere: publishing a price for a region we just routed to a test would invite exactly the mistake this notebook is built to prevent).

One subtlety worth saying out loud — the kind a sharp audience member will probe. The “profit uplift %” is a **posterior mean over every draw**, with the non-elastic draws ($\beta \geq -1$, where raising price doesn’t pay) contributing **0%**. So it is an *expected* uplift already discounted by our uncertainty — not “the uplift once we act”. For the confident **set price** rows ($P(\text{elastic}) \approx 1$) the two coincide; for the weakest region — see the *point-plug illustration* printed under the table — the expected uplift is diluted by the slice of draws worth 0%, and the uplift *conditional on the region truly being elastic* (also printed) is far larger and far shakier.

This is also why you **cannot reproduce the uplift column by plugging the row’s own elasticity into the formula**. The closed form $P^* = c\beta/(\beta + 1)$ blows up as $\beta \rightarrow -1^-$: the printed illustration plugs the weakest region’s *mean* elasticity into the formula and gets an absurd price, a world away from the sane optima of the well-elastic regions. So the per-draw uplift has a heavy right tail and its **mean is not the formula evaluated at the mean elasticity** (Jensen’s inequality, made severe by the boundary). We deliberately propagate the *whole* posterior rather than point-plug a single slope — and that very instability near $\beta = -1$ is the statistical reason those low-confidence regions are routed to a **controlled test**, not to a price move we might regret.

The profit curves make the same point visually. In the *most confident* panel the grey posterior curves agree on a gentle hill with a **flat top**: even a sizeable error in β moves profit little near P^* , so acting is low-risk. In the *least confident* panel the curves fan out and many keep rising off the right edge — the model is begging for a price far outside the green band of prices we have ever charged, on a functional form (A4) we could only test *inside* that band; statistical risk and extrapolation risk stack, so it goes to a test. The *middle* panel sits near the 0.9 action bar and is the boundary case the thresholds are built around: most draws agree on a gentle hill, but a visible minority still explode off to the right — whether it clears the bar is decided by the printed $P(\beta < -1)$, and standing this close to the threshold is itself an argument for measuring before committing. And notice one sobering detail: even for confident regions P^* typically sits *above* the observed price band — the recommendation itself is an extrapolation. That is why a staged rollout (move a minority of stores first, watch a few weeks) beats a fleet-wide overnight change even where the posterior is sure.

0.6.1 The cost assumption — the other number in the formula

$P^* = c\beta/(\beta + 1)$ is **linear in marginal cost c** : a $\pm 25\%$ error in c moves *every* recommended price by $\pm 25\%$, no posterior required. Cost is usually handed to the analyst as a single accounting number, and it deserves the same scepticism as the elasticity — arguably more, because nothing in the data can check it. Below we sweep $c \pm 25\%$ around our €8 and recompute the whole decision layer. Two things to watch: the **action routing never changes** (it keys off $P(\beta < -1)$, which involves no c), but every P^* and every uplift number moves — so if finance restates the cost, the *set-vs-test split* survives while the price list is rewritten wholesale. Cost measurement is as decision-critical as the elasticity itself.

marginal cost €	n	set-price	median P*	range €	fleet	uplift %	uplift 90% CI
6.0		7		14.7-26.3		6	[+2%, +11%]
7.0		7		17.1-30.7		12	[+6%, +19%]
8.0		7		19.6-35.1		21	[+13%, +30%]
9.0		7		22.0-39.5		36	[+25%, +49%]
10.0		7		24.5-43.9		59	[+44%, +76%]

reading it: the set-price roster is c-invariant by construction (column 2 never moves), and every median P* scales exactly linearly in c - get the cost 25% wrong and every recommended price is 25% wrong, with the uplift estimate moving too. Before signing the price list, audit the cost number.

0.7 8 · Caveats — what partial pooling does *not* fix

Partial pooling handles *noise*, not *bias*. If price is endogenous to an **unobserved** demand shock, adjusting for observed Z isn't enough and the elasticity is biased **toward zero** — by step 3's OVB formula, $\hat{\beta} \rightarrow \beta + \kappa \sigma_u^2 / (\kappa^2 \sigma_u^2 + \sigma_v^2)$. We now do three things: **(i)** switch the hidden shock on at the demo strength $\kappa = 1.2$ and refit the full hierarchical model, checking the observed bias against the formula; **(ii)** sweep κ to map the whole bias curve — a range, not an anecdote; **(iii)** deliver the promised **preview of the IV cure**. (These re-runs use the **full** panel, not the thinned subset used above — which only sharpens the fleet estimate and doesn't change the bias point being made.)

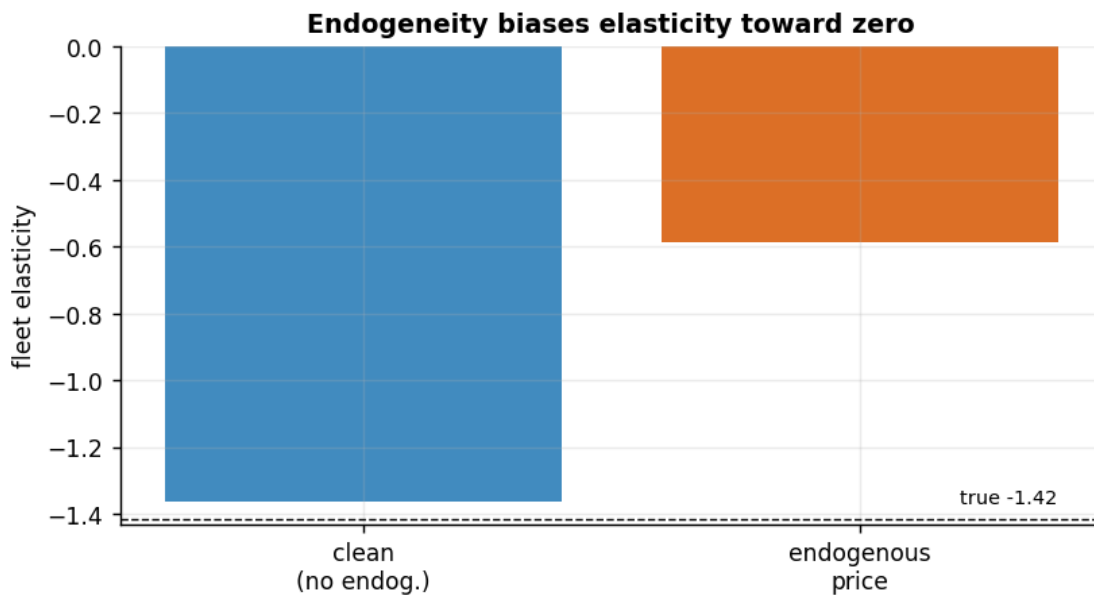
clean fleet elasticity -1.36 (true -1.42)

endogenous fleet elasticity -0.59 (true -1.42) -> biased toward zero by +0.83

OVB formula check - predicted bias +0.83 vs observed +0.83: the bias is computable, not mysterious.

endogenous-fit convergence: max r-hat 1.010 - min ESS 857 - divergences 0

Fix = an instrument (a cost shifter) or a natural price experiment - previewed below, full treatment in notebook 11.

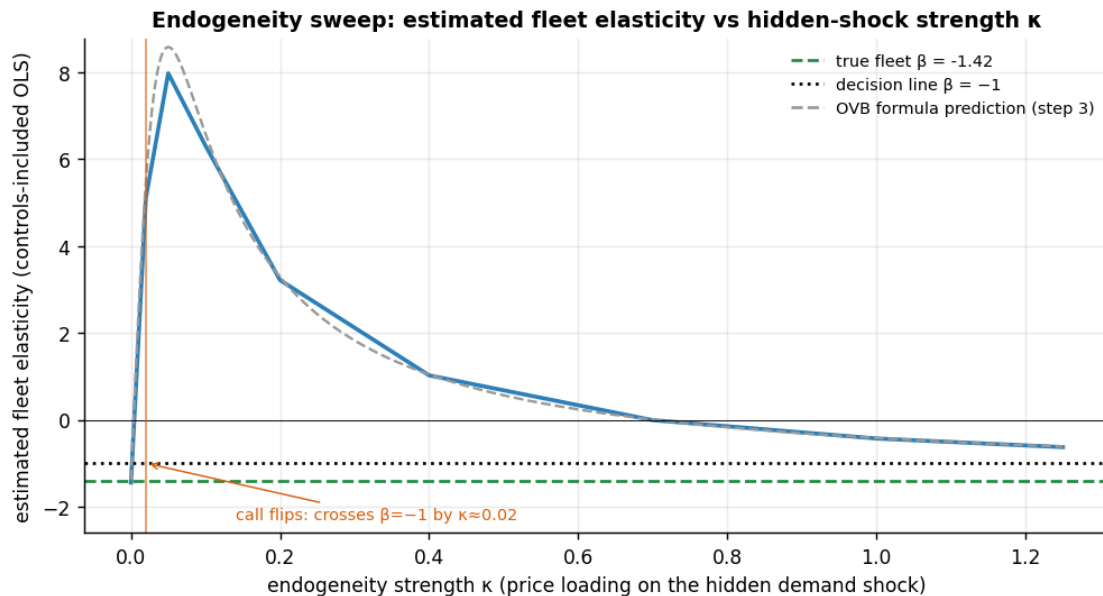


From a point to a curve. One κ is an anecdote. Below we sweep the endogeneity strength from 0 to 1.25 — regenerating the panel and re-estimating the fleet elasticity at each value (a controls-included pooled OLS with region intercepts stands in for the full hierarchical refit: it agrees with it at the $\kappa = 1.2$ point above and costs milliseconds instead of minutes) — and overlay the OVB *prediction* from step 3. Watch two things. (i) The estimate does not drift gently: because the exogenous price wiggle is tiny ($\sigma_\nu = 0.05$), even a faint hidden shock swamps it — the bias term $\kappa\sigma_u^2/(\kappa^2\sigma_u^2 + \sigma_\nu^2)$ peaks near $\kappa \approx \sigma_\nu$ — and the fitted “demand curve” actually slopes *upward* over a whole range of κ . (ii) The recommendation flips almost immediately: the moment the estimate crosses the $\beta = -1$ decision line, “set optimal prices in elastic regions” becomes “demand looks inelastic — hold everywhere”. Endogeneity doesn’t just shade the number; it reverses the pricing call.

estimated fleet elasticity by kappa: kappa=0: -1.45 · kappa=0.02: +5.11 ·
kappa=0.05: +7.98 · kappa=0.1: +6.31 · kappa=0.2: +3.22 · kappa=0.4: +1.02 ·
kappa=0.7: -0.01 · kappa=1: -0.43 · kappa=1.25: -0.63

the estimate is already on the wrong side of $\beta = -1$ by $\kappa \sim 0.02$: a hidden shock explaining only ~14% of residual price variance flips the pricing call fleet-wide.

the grey dashed curve is step 3's OVB formula - theory and refit agree across the whole sweep,
including the perverse region where the fitted demand curve slopes upward.



0.7.1 The cure, previewed — an instrumental variable

The fix for a hidden shock sitting in *both* equations is a variable that moves **price** but has **no path to demand except through price** — an **instrument** W . The textbook pricing instrument is a **cost shifter**: a wholesale-cost or exchange-rate shock moves what we charge, but shoppers never see it except in the price tag. W must satisfy three conditions:

- **Relevance:** $\text{Cov}(W, \log P) \neq 0$ — checked below via the first-stage F-statistic (Stock–Yogo rule of thumb: $F < 10$ = too weak to use, and a weak instrument is *worse* than none);
- **Exclusion:** W affects $\log Q$ *only through* $\log P$ — untestable, argued from institutions (customers don’t observe our wholesale costs);
- **Independence:** $W \perp u$ — cost shocks don’t chase demand shocks.

Two-stage least squares (2SLS) then uses *only the W -driven slice of price variation*:

$$\text{stage 1: } \widehat{\log P} = \hat{\pi}_0 + \hat{\pi}_1 W + \hat{\pi}_Z^\top Z, \quad \text{stage 2: } \log Q = \alpha + \beta_{\text{IV}} \widehat{\log P} + \gamma^\top Z + e.$$

The endogenous slice of price (κu) is orthogonal to W , so stage 1 filters it out and β_{IV} recovers the causal slope. `dgp.price_panel` has no instrument built in, so we plant one on the $\kappa = 1.2$ panel by hand: a cost shock $W_{rt} \sim \mathcal{N}(0, 1)$ that shifts $\log$ -price by $0.25 W_{rt}$ and shifts $\log$ -demand **only through that price move** (by $\beta_r \times$ the price shift) — exclusion true by construction.

first-stage F = 28 (>10: relevant, not weak)

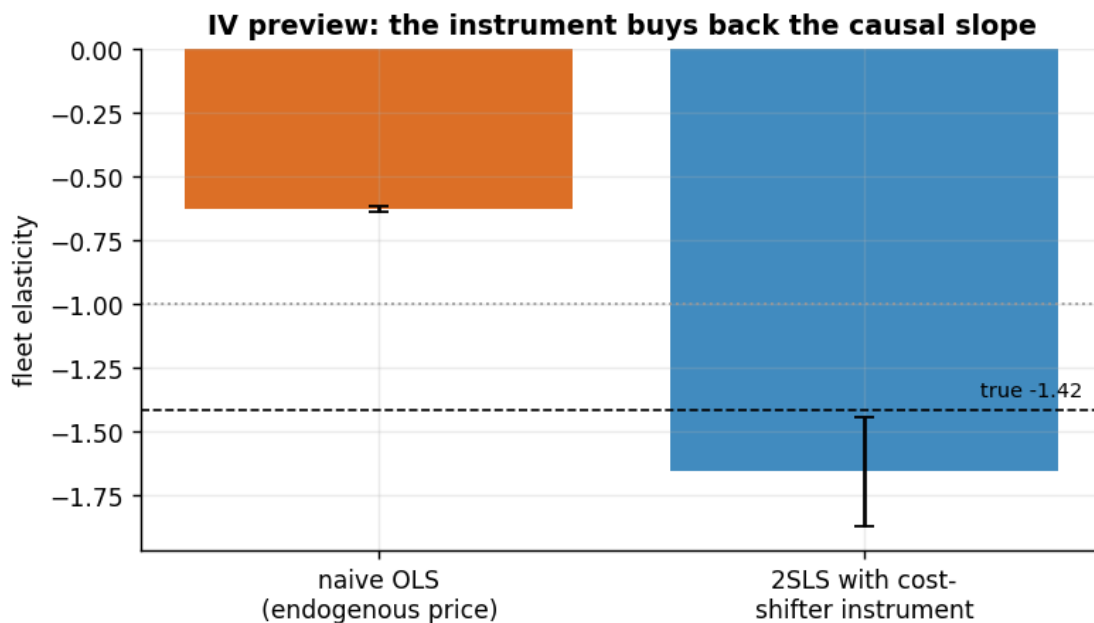
naive OLS: -0.63 ± 0.01 — a tight band 67 SEs from the truth -1.42 (precisely wrong)

2SLS: -1.66 ± 0.21 — a wide band 1.1 SEs from the truth (imprecisely right)

2SLS keeps only the cost-driven slice of price variation, so the hidden-shock

slice drops out of the estimate.

Full treatment - robust SEs, weak-instrument diagnostics, Bayesian IV: notebook 11, and 11b on real Criteo data.



Reading the IV panel. Naive OLS (orange) is **precisely wrong**: a deceptively tight standard error around an estimate dozens of SEs away from the truth — under the table’s routing rules it would have us wrongly *hold* prices fleet-wide, with great confidence. 2SLS (blue) is **imprecisely right**: it lands within a couple of its (much wider) standard errors of the planted elasticity, because it discards every scrap of price variation except the cost-shock slice, which is clean by construction. That asymmetry is the whole IV bargain — the cure costs **variance**: real-world IV estimates are noisier and need bigger samples, and a *weak* instrument (first-stage $F \lesssim 10$) is worse than none, amplifying small exclusion violations into large bias. Managing that trade-off, with honest IV uncertainty in a Bayesian wrapper, is notebook 11’s whole job; notebook 11b repeats it on real Criteo data.

Other caveats, folded into the assumption ledger of step 3.

- **A4 · functional form.** Log-log assumes *constant* elasticity within region. We probed it in-sample (step 5’s curvature term — consistent with zero in this run, with the honest note that $\pm 5\%$ of price wiggle gives that test little power), but every P^* sits at or beyond the edge of the observed price band, where no in-sample check reaches. The staged-rollout advice of step 7 is the operational hedge, and elasticity should be re-estimated as the new prices generate data.
- **A3 · dynamics.** A $\text{lag}(\text{price})$ term would capture stockpiling / reference-price dynamics (omitted here for clarity); on real scanner data always try it — pantry-loading masquerades as high short-run elasticity and vanishes at longer horizons.
- **Levels vs logs — a porting trap.** The model is for $\log Q$; if you ever forecast *units*, the lognormal mean needs the correction $\mathbb{E}[Q | P] = e^{\mu + \sigma^2/2}$, not e^μ . Our profit *uplift* is immune

— it is a **ratio** of two expected profits with the same residual σ , so the $e^{\sigma^2/2}$ factor cancels in `uplift_pct` — but the moment you report absolute euros or units, it bites.

- **Pooling is a prior choice.** Too few regions or a mis-specified τ_β prior can over-pool — always inspect the fleet-vs-region spread (τ_β 's posterior), and remember step 4's non-centered reparametrization when the sampler starts complaining.

0.8 9 · The one-paragraph decision

To the CMO. Fleet demand is elastic — the fleet-mean elasticity lands close to -1.4 (exact value and credible interval in the dict below), with genuine region-to-region spread, so a single national price is leaving money on the table. For the majority of regions the posterior clears our 0.9 bar that demand is truly elastic ($\beta < -1$); for those we recommend moving toward the table's **P*** prices — the expected fleet profit uplift and its 90% interval are printed under the decision table — rolled out to a minority of stores first, because every P* sits at or beyond the price range we have ever observed. The remaining regions get **no price move**: their elasticity posterior straddles -1 , where the optimal-price formula is unstable, so they get a **controlled price test** — randomized $\pm 10\%$ price cells for 8 weeks, deliberately manufacturing the exogenous price variation (A2) that identification feeds on. Three conditionals we will not hide: **(i)** everything assumes prices were not historically managed against demand (A1) — if they were, step 8 says the true elasticity is *more negative* than we estimated, so the uplift is conservative on that axis; **(ii)** the constant-elasticity extrapolation (A4) cuts the other way — profit at prices we never charged is model, not measurement; **(iii)** the recommended prices scale one-for-one with the €8 marginal-cost figure, so a cost restatement rewrites the price list (though not the set-vs-test split). Act where the posterior is sure; measure where it is not.

```
{
  "fleet_elasticity": -1.36,
  "fleet_elasticity_ci90": [-1.53, -1.16],
  "tau_beta": 0.31,
  "coverage90_true_beta": 0.67,
  "n_regions": 12,
  "n_set_price": 7,
  "n_controlled_test": 5,
  "n_hold": 0,
  "fleet_uplift_pct": 21.0,
  "fleet_uplift_ci90_pct": [13.1, 30.4],
  "marginal_cost_eur": 8.0,
  "endogeneity_bias_at_kappa_1p2": 0.83,
  "ovb_predicted_bias_at_kappa_1p2": 0.83,
  "kappa_where_call_flips": 0.02,
  "iv_first_stage_F": 28.5,
  "beta_ols_endogenous": -0.63,
  "beta_iv": -1.66,
  "beta_iv_se": 0.21,
  "true_fleet_elasticity_reference": -1.42
}
```